

****TECHNICAL AND COMMERCIAL PROPOSAL****

****1. EXECUTIVE SUMMARY****

The Co-operative Development Section of the Embu County Government plays a critical role in driving the economic development of the county, overseeing approximately 200 cooperatives across vital value chains, including Coffee, Dairy, Miraa, and Irrigation. Managing an ecosystem of this size manually or via distributed spreadsheets presents severe operational challenges in data integrity, regulatory oversight, staff accountability, and document security.

We propose the development of a secure, enterprise-grade, centralized ****Cooperative Management and Governance System****. This custom web-based application will replace manual processes with a streamlined digital environment. The platform will serve three primary user groups: County Directors, Field and Sub-County Staff, and Cooperative Managers.

By implementing this system, Embu County will establish an immutable repository of cooperative memberships, automate field staff reporting, digitize governance compliance (such as the mandatory 1/3 gender and leadership rotation rules), and secure sensitive legal documentation via a strict, multi-tier administrative approval conveyor belt.

****2. PROBLEM STATEMENT ***** PROJECT OBJECTIVES****

****2.1 Current Operational Challenges****

- ****Information Asymmetry:**** Lack of a single source of truth for cooperative profiles, membership registries, and financial share capital allocations.
- ****Oversight Gaps:**** Difficulty tracking field officer visits, leave tracking, and ground-level achievements in real time.
- ****Compliance Risks:**** Manual tracking increases the risk of cooperatives operating with expired leadership committees, non-compliant governance structures, or unvetted statutory documents.
- ****Document Vulnerability:**** Critical files like By-Laws, Audit Reports, and AGM Minutes are vulnerable to damage or loss when stored physically.

****2.2 Core Project Objectives****

- ****Centralize Records:**** Create a secure digital registry for all 200 cooperatives, categorized by value chain and geographic location (Sub-county/Ward).
- ****Automate Workflows:**** Implement a strict digital chain of command where submitted cooperative files are systematically verified by Sub-county officers and approved by the Director before publication.
- ****Enforce Compliance:**** Use backend algorithms to automatically track committee election cycles, term expirations, and statutory requirements (e.g., the 1/3 rotation rule).
- ****Enhance Field Efficiency:**** Provide a structured portal for county staff to schedule weekly visits, obtain management approvals, and submit post-visit achievement logs.

****3. COMPREHENSIVE SYSTEM SCOPE ***** FUNCTIONAL MODULES****

The system will be explicitly developed around five core functional modules based on the statutory and operational brief provided by the Directorate:

****Module 1: County Staff ***** Access Management (RBAC)****

- ****Role-Based Access Control:**** A secure administrative panel allowing the County Director to create, modify, or revoke staff user accounts and dynamically assign granular access rights to individual tabs and modules.

- ****Staff Profiles:**** Comprehensive digital identities capturing Key metrics: Full Name, Job Group, Official Designation, Phone Number, Corporate Email, Reporting Structure, and assigned Office Location (Sub-county and Ward drop-downs).

****Module 2: Field Operations ****&**** Leave Tracking****

- ****Leave Management Portal:**** An integrated, self-service utility for staff to apply for, track, and process leave requests within the department.

- ****Field Visit Planner:**** A proactive scheduling tracker where field officers must log their weekly planned cooperative visits. The system routes these plans to the next-in-line manager for digital authorization.

- ****Post-Visit Reporting:**** An mandatory interface for officers to input narrative feedback, achievements, data points, and immediate next-action items following a field engagement.

****Module 3: Cooperative Registry ****&**** Member Database****

- ****Cooperative Profiles:**** A master directory of all 200 cooperatives, fully searchable and filterable by specific Value Chains (Coffee, Dairy, Miraa, Irrigation, etc.) and mapped directly to their respective geographic Wards.

- ****Digital Member Roll:**** An isolated, secure database within each cooperative profile tracking individual member analytics: Legal Name, National Identity (ID) Number, Active Phone Numbers, Gender/Sex, and allocated Share Capital.

****Module 4: Secure Document Management System (DMS)****

- ****Encrypted PDF Repository:**** Dedicated cloud storage interfaces optimized for uploading, archiving, and cataloging official legal and regulatory documents, specifically:

- Cooperative By-Laws

- Dated Meeting Minutes

- Corporate Code of Conduct Documents

- Official Audit Reports

- Departmental Spot Check Reports

- ****Multi-Tier Approval Conveyor Belt:**** An un-bypassable state engine ensuring that any document uploaded by a cooperative manager remains in a "Pending" quarantine state until reviewed and verified by the assigned Sub-county officer, and officially signed off by the County Director.

****Module 5: Governance, Election ****&**** AGM Tracking****

- ****Election Lifecycle Engine:**** Automated tracking modules capturing the exact date a committee member is elected and dynamically calculating their precise re-election due date.

- ****Governance Compliance Auditing:**** Algorithmic verification mechanisms checking committee submissions against statutory rules, explicitly flagging compliance or non-compliance regarding the ****1/3 leadership rotation/gender rule****.

- **Candidate Application Interface:** A digital application pipeline allowing members to apply for elections directly through the platform, with assisted-entry features managed by cooperative managers or county field staff.
- **Committee Structural Mapping:** Dedicated management interfaces for distinct governing organs:
- **Management Committee:** Separate data tracking for Executive Roles (Chairperson, Vice Chairperson, Honorable Secretary, Treasurer), ordinary Board Members (4 slots), and the non-elected Executive Manager.
- **Signatories Tab:** A highly restricted, auditable data sheet displaying the formally approved bank account signatories for the cooperative.
- **Supervisory Committee:** Data capturing for the Chair and Secretary roles.
- **AGM Archival Module:** A robust interface dedicated to Annual General Meetings and Extraordinary AGMs, capturing drop-down classifications, official Letters Calling for the AGM, exact meeting dates, and signed Minutes.

4. SYSTEM ARCHITECTURE & TECHNICAL SPECIFICATIONS

4.1 Technical Stack Specification

- **Frontend Architecture:** Built using **React.js** and **Next.js**. This layout ensures optimized server-side rendering, exceptional client-side state execution for large-scale data tables, fast interface loading speeds across standard connections, and mobile-responsive UI frameworks.
- **Backend Runtime Environment:** Powered by a high-throughput **Node.js** environment. The application programming interfaces (APIs) are exposed as a secure RESTful layer handling data transport using encrypted JSON payloads.
- **Database Management System:** Utilizing **PostgreSQL**, an enterprise-class object-relational database system. PostgreSQL guarantees absolute transactional data integrity (ACID compliance), which is vital for securing national identity records, financial share capital structures, and multi-user audit logs.
- **File Storage Layer:** Static files and binary items (PDF documents) are entirely decoupled from the application database and isolated within a secure, encrypted Cloud Object Storage repository. Files are delivered to authorized clients on-demand via time-restricted, cryptographically signed pre-signed uniform resource locators (URLs).

5. PROJECT IMPLEMENTATION TIMELINE

The development lifecycle is structured over an intensive 12-to-14-week phase, ensuring proper testing, revision loops, and milestone evaluations.

Phase	Milestone Phase Description	Key Deliverables
Duration		
---	---	---
Phase 1	System Design, Schema Layout & RBAC Setup	Technical Specifications Document; UI/UX interactive wireframe designs; Database schema provisioning; Module 1 & 2 launch. Weeks 1 – 3
Phase 2	Core System Development & DMS Integration	Full building of Cooperative Directories, Member Databases, and the Multi-tier Document Approval Pipeline (Modules 3 & 4). Weeks 4 – 7
Phase 3	Governance Engine & Compliance Logic	Implementation of Election tracking algorithms, 1/3 rotation checking mechanisms, and AGM filing tools (Module 5). Weeks 8 – 10
Phase 4	Quality Assurance, Testing & Optimization	Comprehensive end-to-end security audits, data sanitization testing, validation loops, and

performance optimization. | Weeks 11 – 12 |
| ****Phase 5**** | Data Migration, Training & Handover | Execution of bulk Excel/CSV member list imports; On-site user training workshops for county staff; Official system handover. | Weeks 13 – 14 |

****6. FINANCIAL PROPOSAL ****&**** PAYMENT MILESTONES****

****6.1 Cost Breakdown****

The project budget is presented as a fixed-price contract covering comprehensive system architecture, construction, and training components:

Item Code	**Deliverable Module / Implementation Component**	**Cost (KES)**
EMP-001	System Design Architecture, UI/UX Prototypes & Secure Database Provisions	250,000
EMP-002	Core Software Development (Modules 1, 2, 3 & 5 Full Implementations)	950,000
EMP-003	Document Management System & Multi-Tier File Approval Pipeline Setup	300,000
EMP-004	Cloud Systems Deployment, Data Migration Scripts & User Training Workshops	200,000
TOTAL	**Total Project Development Capital (Fixed Cost)**	**1,700,000 KES**

****6.2 Project Payment Milestones****

To protect both parties and maintain momentum, funding is aligned to explicit system delivery indicators:

- ****Milestone 1 (Mobilization Fee): 30% of Total Cost (510,000 KES)**** due upon contract execution and prior to the initiation of UI/UX system design wireframes.
- ****Milestone 2 (Mid-Point Deliverable): 30% of Total Cost (510,000 KES)**** due upon completion of Phase 2 (successful deployment of the Staff Portal, Cooperative Directory, and operational Document Upload Pipeline).
- ****Milestone 3 (Functional System Complete): 20% of Total Cost (340,000 KES)**** due upon successful completion of Phase 3 (Integration of Governance Engine, Election logs, and implementation to a staging/testing environment).
- ****Milestone 4 (Final Handover ****&**** Sign-Off): 20% of Total Cost (340,000 KES)**** due upon final User Acceptance Testing (UAT) sign-off, completion of staff training workshops, and deployment to the official live county platform.

****6.3 Recurring Infrastructure ****&**** Maintenance Framework****

(Commencing 12 months after system launch)

- ****Annual System Maintenance ****&**** Technical Support:**** Billed at a standard ****20% of the core software construction cost per annum****. This ensures ongoing optimization, data patch adjustments, routine package updates, and dedicated technical helpdesk priority access.
- ****Production Cloud Infrastructure/Hosting:**** Billed directly to the County at pass-through cost based on server utilization (e.g., memory, bandwidth, block storage consumption for PDF files).

****7. TERMS, CONDITIONS ****&**** PROJECT GOVERNANCE****

****7.1 County Responsibilities ****&**** Dependency Scope****

- **Clean Data Provision:** Embu County Government assumes full responsibility for collecting and organizing existing cooperative membership records into clean, standardized digital formats (Excel/CSV templates provided by the developer) to enable automated bulk-importing scripts.

- **IT Liaison Appointment:** The county will provide a dedicated point-of-contact within the IT or Cooperative department to manage infrastructure access, domain mapping, and coordinate scheduling for validation reviews and user testing.

7.2 Source Code Ownership & Intellectual Property

- **Full IP Transfer:** Upon the final payment milestone execution, the complete customized source code, relational database layouts, design schemas, and intellectual property items will be fully transferred to the Embu County Government.

- **Zero Proprietary Licensing Lock-in:** The software architecture leverages modern open-source framework ecosystems. The County will never be subject to ongoing developer licensing costs, seat metrics, or proprietary system fees.

7.3 Professional Warranty & System Stabilization

- **90-Day Stabilization Warranty:** The platform is covered by a 3-month comprehensive software warranty starting immediately upon system launch. This warranty covers the swift correction of unexpected errors, edge-case bug fixes, or minor visual adjustments at no additional cost to the department.